# Analytic Number Theory

Junchi Wang

*jw3724@ic.ac.uk*

Based on course notes by *Ian Petrow*

# Contents

# Notation

This section summarizes the main symbols and notations used throughout the thesis.

| **Symbol** | **Meaning** |
|---|---|
| $d\|n$ | d is a divisor of n |
| $\lceil\cdot\rceil$ | ceil function |
| $\lfloor\cdot\rfloor$ | floor function |
| $f = O(g)$ | f is bounded by g, i.e. $\|f(x)\| \leq Cg(x)$ |
| $f \sim g$ | $\frac{f(x)}{g(x)} \to 1$ |
| $n!$ | $\prod_{1\leq k\leq n} k$ |
| $\binom{a}{b}$ | binomial, i.e.$\frac{a!}{(a-b)!b!}$ |
| $(m, n)$ | greatest common divisor (gcd) of $m, n$ |
| $[m, n]$ | least common multiple (lcm) of $m, n$ |
| $a \equiv b \pmod q$ | a and b are congruent modulo q |
| $f : x \mapsto f(x)$ | define function $f$ and variable $x$ |
| $\mathbb{Z}^+$ or $(\mathbb{Z}, +)$ | group of integer set under addition operation |
| $\hat{f}$ | Fourier Transform of $f$ |
| $\tilde{f}$ | Mellin Transform of $f$ |

# Function

This section summarizes the main symbols and notations used throughout the thesis.

| Symbol | Meaning |
|---|---|
| $\pi(x)$ | prime counting function (1.1) |
| $\zeta(s)$ | zeta function (1.2) |
| $\Lambda(n)$ | von Mangoldt function (2.1) |
| $M_f(x)$ | summation function of $f$ (2.2) |
| $\mu(n)$ | Möbius function (2.5) |
| $L(s, f)$ | Dirichlet Series (3.2) |
| $\varphi(n)$ | Euler function (4.3) |
| $e(x)$ | expontential function (6.1) |
| $\Gamma(s)$ | Gamma function (6.4) |

# Mathematics: How it Works

Mathematics is like building a castle from bricks. We usually start from *definition*, then prove some *proposition* and *lemma*, and finally prove the central results *theorem* and its consequence *corollary*.

| **Type** | **Purpose** | **Typical Usage** |
|---|---|---|
| Definition | Introduces a new concept, object, or notation. | Used when explaining what something *is*. |
| Proposition | A less central but still important result. | Often a useful or supporting fact. |
| Lemma | A smaller, helper result used to prove a theorem. | Usually not important by itself, but essential in proofs. |
| Theorem | A major mathematical statement that has been proved. | Central results of a section or paper. |
| Corollary | A direct consequence of a theorem or proposition. | Follows almost immediately from a previous result. |

**Note:** The reader is assumed to have a basic background in *group theory* and *complex analysis*.

# 1 Counting Prime Numbers

## 1.1 Introduction

It has been known since the time of Euclid that there are infinitely many prime numbers. Arguing by contradiction, suppose that there were only finitely many primes $p_1, \ldots, p_n$. Then the number $p_1 \cdots p_n + 1$ must have a prime divisor not equal to any of $p_1, \ldots, p_n$. In this course we will be interested in quantifying the infinitude of prime numbers. To do so, we define the prime counting function

$$\pi(x) = \#\{\, p \in \mathcal{P} : p \leq x \,\}.$$

Euclid's theorem therefore says that $\pi(x) \to \infty$ as $x \to \infty$, but the question is

*at what rate?*

**Theorem 1.1** (Prime Number Theorem (PNT))**.** *As $x \to \infty$ we have*

$$\pi(x) \sim \frac{x}{\log x}.$$

### Remark 1.3.

In the 1830s, Dirichlet formulated the prime number conjecture in a slightly different form: introducing the following function, called the *logarithmic integral*

$$\mathrm{Li}(x) = \int_2^x \frac{dt}{\log t},$$

he conjectured that

$$\pi(x) \sim \mathrm{Li}(x).$$

Seeing as $\mathrm{Li}(x) \sim \dfrac{x}{\log x}$ (do an integration by parts) this formulation of the conjecture is equivalent to the original. However, Dirichlet's version is better: as we will see later, the proof of the PNT in fact gives the following asymptotic formula

$$\pi(x) = \mathrm{Li}(x) + O\big(x \exp(-c\sqrt{\log x})\big)$$

for some absolute constant $c > 0$. The celebrated *Riemann Hypothesis* predicts that

$$\pi(x) = \mathrm{Li}(x) + O\big(x^{1/2}(\log x)^2\big).$$

## 1.2 Euler's Method

**Zeta Function:** For $s > 1$ one considers the convergent series

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s}.$$

In terms of prime numbers:

$$\zeta_p(s) = 1 + \frac{1}{p^s} + \frac{1}{(p^2)^s} + \cdots + \frac{1}{(p^\alpha)^s} + \ldots.$$

As geometric series, we have

$$\zeta_p(s) = (1 - 1/p^s)^{-1}$$

**Key observation:**

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s} = \prod_p \zeta_p(s) = \prod_p (1 - 1/p^s)^{-1} \tag{1}$$

**Theorem 1.2** (Fundamental Theorem of Arithmetic)**.** *Every integer $n \geq 2$ can be expressed as a product of prime numbers. Moreover, this factorization is unique up to the order of the factors. That is, there exist distinct prime numbers $p_1, p_2, \ldots, p_k$ and positive integers $\alpha_1, \alpha_2, \ldots, \alpha_k$ such that*

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}.$$

Assume there are finite primes, $\prod_p (1 - 1/p^s)^{-1}$ will be finite, but the $\sum_{n \geq 1} \frac{1}{n^s}$ will be infinite as $s \to 1$, which contradicts our assumption, thus the number of primes is infinite.

If we take logarithm on each side, the equation can be written as,

$$\log \zeta(s) = \log \prod_p (1 - 1/p^s)^{-1} = -\sum_p \log(1 - 1/p^s) \approx \sum_p 1/p^s \tag{2}$$

The $\zeta(s)$ is infinite as $s \to 1$, thus the series

$$\sum_p 1/p \tag{3}$$

is *divergent.*

## 1.3 Chebyshev's Method

We begin with the follow result of Chebyshev (circa 1850), which only uses elementary methods and which gives the correct order of magnitude for the function $\pi(x)$.

**Theorem 1.3.** *There exist constants $0 < c < C$ such that for $x \geq 2$ one has*

$$c\frac{x}{\log x} \leq \pi(x) \leq C\frac{x}{\log x}.$$

**Definition 1.4** ($p$-adic valuation)**.** *For $n \in \mathbb{Z} \setminus \{0\}$ and $p$ a prime number, the $p$-adic valuation of $n$, written $v_p(n)$, is the largest integer $\alpha \geq 0$ such that $p^\alpha$ divides $n$. That is to say, such that $p^\alpha \mid n$ and $p^{\alpha+1} \nmid n$. In particular, one has*

$$n = \prod_{p|n} p^{v_p(n)} = \prod_{p \in \mathcal{P}} p^{v_p(n)}.$$

Define $\theta(x)$:

$$\theta(x) = \sum_{p \leq x} \log p$$

Consider binomial coefficient $\binom{2n}{n} = \frac{2n!}{(n!)^2}$

$$\log \binom{2n}{n} = \sum_{p \leq 2n} v_p \binom{2n}{n} \log p$$

With some care, the binomial contains all prime divisor between $n$ and $2n$.

$$\sum_{p \leq 2n} v_p \binom{2n}{n} \log p \geq \sum_{n \leq p \leq 2n} \log p = \theta(2n) - \theta(n)$$

On the other hand, by approximating by integrals,

$$\log n! = \sum_{i=1}^{n} \log i = \int_1^n \log t \, dt + \Theta(\log n - \log 1) = n \log n - n + O(\log n).$$

$$\log \binom{2n}{n} = 2n \log 2 + O(\log n)$$

We can get,

$$\theta(2x) - \theta(x) \leq 2x \log(2) + O(\log x)$$

By telescoping summation, we have,

$$\theta(x) = \sum_{k \geq 0} \theta(x/2^k) - \theta(x/2^{k+1}) \leq \sum_{k \geq 0} x/2^k \log(2) + O(\log(x)) = 2x \log(2) + O(\log(x))$$

Finally,

$$\pi(x) = \sum_{p \leq x} 1 = \sum_{p \leq x^{1/2}} 1 + \sum_{x^{1/2} \leq p \leq x} 1$$

The first term is bounded $x^{1/2}$, the second term,

$$\sum_{x^{1/2}\le p\le x} 1 \le \frac{1}{\log x^{1/2}} \sum_{x^{1/2}\le p\le x} \log p = \frac{2}{\log x}(\theta(x) - \theta(x^{1/2})) = 4\log 2\frac{x}{\log x} + O(x^{1/2})$$

The **upper bound**,

$$\pi(x) \le 4\log 2\frac{x}{\log x} + O(x^{1/2}) \tag{4}$$

To get a lower bound, take a look at binomial,

$$v_p\binom{2n}{n} = v_p(2n!) - 2v_p(n!)$$

By changing the order of integrals,

$$v_p(2n!) - 2v_p(n!) = \sum_{\alpha\ge 1}\lfloor\frac{2n}{p^\alpha}\rfloor - 2\lfloor\frac{n}{p^\alpha}\rfloor = \sum_{\alpha\ge 1}\omega(\frac{n}{p^\alpha})$$

where

$$\omega(x) = \lfloor 2x\rfloor - 2\lfloor x\rfloor = \begin{cases} 0 & x \mod 1 \in [0, 1/2] \\ 1 & x \mod 1 \in [1/2, 1] \end{cases}$$

Note that $\omega(x) \le 1$ and $\alpha \le \log 2n/\log p$,

$$\log\binom{2n}{n} = 2n\log 2 + O(\log n) = \sum_{p\le 2n} v_p\binom{2n}{n}\log p \le \sum_{p\le 2n}\log 2n/\log p\log p = \pi(2n)\log(2n)$$

Finally, we have **lower bound**,

$$\pi(n) \ge \log 2\frac{n}{\log n}$$

**Theorem 1.5** (Mertens). *We have*

$$\sum_{p\le x}\frac{\log p}{p} = \log x + O(1).$$

*Proof.* **Key Observation:**

$$n! = \prod_{p\le n} p^{v_p(n!)} \tag{5}$$

$$\log(n!) = \sum_{p\le n} v_p(n!)\log p$$

The left hand side can be written as,

$$\log(n!) = \sum_{1\le x\le n}\log x \approx \int_1^n \log x \mathrm{d}x = n\log(n) + O(n)$$

The right hand side can be written as

$$\sum_{p\leq n} v_p(n!)\log p = \sum_{p\leq n}\log p \sum_{x\leq n}\sum_{a\geq 1, p^a|x} 1 = \sum_{p\leq n}\log p \sum_{a\geq 1}\sum_{x\leq n, p^a|x} 1$$

Which can be expressed as,

$$= \sum_{p\leq n}\log p \sum_{a\geq 1}\lfloor\frac{n}{p^a}\rfloor = \sum_{p\leq n}\log p\frac{n}{p} + O(n)$$

Finally we have,

$$\sum_{p\leq n}\log p\frac{n}{p} + O(n) = n\log(n) + O(n)$$

thus,

$$\sum_{p\leq n}\frac{\log p}{p} = \log(n) + O(1) \tag{6}$$

$\square$

# 2 Sums of arithmetic functions

## 2.1 arithmetic functions

**Definition 2.1.** *An arithmetic function is a complex-valued function on the positive integers, $f : \mathbb{N}_{\geq 1} \to \mathbb{C}$. We write $\mathcal{A}$ for the $\mathbb{C}$-vector space of arithmetic functions.*

The *von Mangoldt* function

$$\Lambda(n) = \begin{cases} \log p, & n = p^\alpha,\ \alpha \geq 1, \\ 0, & n \neq p^\alpha. \end{cases}$$

**Definition 2.2.** *Let $f$ be an arithmetic function. The summation function of $f$ is the function defined on $\mathbb{R}_{\geq 0}$ by*

$$x \mapsto M_f(x) = \sum_{1 \leq n \leq x} f(n).$$

The summation function of $f$ is a piecewise constant function, and in this chapter, we will present methods to study the following question:

## 2.2 Approximation by integrals

If $f$ is the restriction to $\mathbb{N}_{\geq 1}$ of a continuous function on $\mathbb{R}$, then $M_f(x)$ is often well approximated by

$$\int_1^x f(t)\, dt.$$

For example, if $f$ is *monotone* we have

**Theorem 2.3.** *Monotone comparison If $f$ is monotone we have*

$$M_f(x) = \int_1^x f(t)\, dt + O(f(x) - f(1)). \tag{7}$$

One can use this result to estimate $\log n!$. Indeed, setting $f(n) = \log n$, one obtains

$$\log n! = \sum_{i=1}^{n} \log i = \int_1^n \log t\, dt + \Theta(\log n - \log 1) = n \log n - n + O(\log n).$$

thus providing a fairly good approximation for $\log n!$.

**Theorem 2.4** (Integration by parts)**.** *Let $g$ be an arithmetic function. Let $a < b \in \mathbb{R}_{>0}$ and $f : [a,b] \to \mathbb{C}$ a $C^1((a,b])$ function. We have*

$$M_{fg}(b) - M_{fg}(a) = \sum_{a<n\leq b} f(n)g(n) = [f(t)M_g(t)]_{t=a}^{t=b} - \int_a^b M_g(t)f'(t)\,dt$$

$$= f(b)M_g(b) - f(a)M_g(a) - \int_a^b M_g(t)f'(t)\,dt.$$

## 2.3 Dirichlet convolution

The Dirichlet convolution is a composition law on the set of arithmetic functions that realizes the multiplicative structure of the integers.

Let $f, g \in \mathcal{A}$, and define $f * g \in \mathcal{A}$ by setting

$$(f * g)(n) = \sum_{ab=n} f(a)g(b) = \sum_{d|n} f(d)g(n/d)\,.$$

**Example:**

$$\log = \Lambda * 1, \quad \text{i.e.} \quad \log(n) = \sum_{d|n} \Lambda(d).$$

Indeed, if $n = \prod_p p^{\alpha_p}$ then

$$\begin{aligned}\log(n) &= \log\left(\prod_p p^{\alpha_p}\right)\\ &= \sum_p \alpha_p \log(p)\\ &= \sum_p \sum_{1\leq\alpha\leq\alpha_p} \log(p)\\ &= \sum_{p^\alpha|n} \log(p) = \sum_{d|n} \Lambda(d).\end{aligned}$$

**Möbius inversion formula:**

**Definition 2.5.** *The* Möbius function *is by definition the inverse of the constant function 1:*

$$\mu = 1^{(-1)}, \quad \mu(1) = 1, \quad \mu(n) = -\sum_{\substack{d|n\\ d<n}} \mu(d) \quad \text{for } n \geq 2.$$

In the following section we will show that

1. If $n$ is divisible by a square not equal to 1 (i.e. there exists a prime $p$ such that $p^2 \mid n$), then $\mu(n) = 0$.

2. If $n$ is square-free, and has $r$ prime factors (i.e. $n = p_1 \cdots p_r$), then $\mu(n) = (-1)^r$.

The inverse of the constant function 1 indicates that

$$\mu * 1 = \delta$$

where $\delta(n) = \begin{cases} 1 & n = 1 \\ 0 & n > 1 \end{cases}$

*Proof.* when $n > 1$,

$$\begin{aligned} \mu * 1(n) &= \sum_{d|n} \mu(d) \\ &= \sum_{d|n, p^2|d} \mu(d) + \sum_{d|n, p^2 \nmid d} \mu(d) \\ &= 0 + \sum_{d|n, p^2 \nmid d} \binom{r}{x} (-1)^x 1^{r-x} \\ &= 0 \end{aligned}$$

□

## 2.4 Applications to Counting Prime Numbers

**Theorem 2.6** (Mertens). *We have*

$$\sum_{n \le x} \frac{\Lambda(n)}{n} = \log(x) + O(1). \tag{8}$$

*Proof.* start from

$$\sum_{n \le x} \log n = x \log x + O(x) \tag{9}$$

The left hand side can be written as,

$$\sum_{n \le x} \log n = \sum_{n \le x} \sum_{d|n} \Lambda(d) = \sum_{d \le x} \lfloor \frac{x}{d} \rfloor \Lambda(d) = \sum_{d \le x} \frac{x}{d} \Lambda(d) + O(x)$$

Finally we have,

$$\sum_{d \le x} \frac{x}{d} \Lambda(d) + O(x) = x \log x + O(x) \tag{10}$$

□

## 2.5 Multiplicative functions

**Definition 2.7.** *A non-zero arithmetic function $f$ is called multiplicative if and only if for all $m, n \geq 1$ with $(m, n) = 1$ we have $f(mn) = f(m)f(n)$. A non-zero arithmetic function is called completely multiplicative if for all $m, n \geq 1$ we have $f(mn) = f(m)f(n)$.*

**Proposition 2.8.** *If $f$ and $g$ are multiplicative, then $f * g$ and $f^{(-1)}$ are as well.*

# 3 Dirichlet Series

## 3.1 Review of Power Series

For a sequence $a_n$, the power series is defined as,

$$F(a,q) = \sum_{n\geq 0} a_n q^n \tag{11}$$

The radius of convergence is defined as,

$$\frac{1}{\rho} = \limsup_{n\to\infty} |a_n|^{1/n} \tag{12}$$

Let $b_n$ be another sequence with associated power series,

$$F(b,q) = \sum_{n\geq 0} b_n q^n$$

The product will be

$$F(a,q)F(b,q) = \sum_{n\geq 0} c_n q^n \quad c_n = \sum_{k+l=n} a_k b_l \tag{13}$$

## 3.2 Dirichlet Series

Dirichlet series are to arithmetic functions as power series are to sequences of numbers. Let $f \in \mathcal{A}$ be an arithmetic function. The *Dirichlet series* associated to $f$ is the series in the complex variable $s$ given by

$$s \mapsto L(s,f) = \sum_{n\geq 1} \frac{f(n)}{n^s}.$$

**Definition 3.1.** *An arithmetic function $f : \mathbb{N}_{\geq 1} \to \mathbb{C}$ is of* polynomial growth *if it satisfies one of the following equivalent conditions.*

- *There exists a constant $A \in \mathbb{R}$ (depending on $f$) such that $|f(n)| = O(n^A)$.*
- *There exists $\sigma \in \mathbb{R}$ such that the series $L(\sigma, f)$ is absolutely convergent.*

In this case we write

$$\sigma_f = \inf\{\sigma \in \mathbb{R} : L(\sigma, f) \text{ converges absolutely}\} \in \mathbb{R} \cup \{-\infty\};$$

The number $\sigma_f$ is called the *abscissa of convergence* of $L(s,f)$.

*Proof.* Exercise. □

**Proposition 3.2.** *Let $f$ be an arithmetic function with polynomial growth, and let $\sigma_f$ be its abscissa of convergence. For all $\sigma > \sigma_f$, the series $L(s, f)$ converges absolutely and uniformly in the half-plane $\{s \in \mathbb{C} : \mathrm{Re}(s) \geq \sigma\}$. In this domain, the derivative of $L(s, f)$ is the Dirichlet series of the arithmetic function*

$$-\log f : n \mapsto -\log(n)f(n),$$

*that is to say,*

$$L'(s, f) = L(s, -\log f) = \sum_{n\geq 1} \frac{-\log(n)f(n)}{n^s},$$

*which has abscissa of convergence $\sigma_f$ as well.*

*Proof.* To prove abscissa of convergence $\sigma_{-\log . f} = \sigma_f$, for $n \geq 3$,

$$\log n|f(n)| > |f(n)| \tag{14}$$

which indicates that $\sigma_{-\log . f} \geq \sigma_f$ On the other hand,

$$L'(s, f) = L(s, -\log f) = \sum_{n\geq 1} \frac{-\log(n)f(n)}{n^s},$$

which converges on $\mathrm{Re}(s) \geq \sigma_f$, the function $-\log . f$ have a convergence area greater or equal than $\mathrm{Re}(s) \geq \sigma_f$, i.e.

$$\sigma_{-\log . f} \leq \sigma_f$$

Thus we have $\sigma_{-\log . f} = \sigma_f$ □

The main reason to introduce Dirichlet series is the following.

**Theorem 3.3.** *Let $f, g \in \mathcal{A}$, with $\sigma_f, \sigma_g < \infty$. Then, $\sigma_{f*g} \leq \max(\sigma_f, \sigma_g)$, and for $\mathrm{Re}(s) > \max(\sigma_f, \sigma_g)$ we have*

$$L(s, f * g) = L(s, f)L(s, g).$$

*Proof.* Let $\mathrm{Re}(s) > \max(\sigma_f, \sigma_g)$, so that

$$\sum_{n=1}^{\infty} \frac{|f * g(n)|}{|n^s|} = \sum_{n=1}^{\infty} \frac{|\sum_{ab=n} f(a)g(b)|}{n^{\mathrm{Re}(s)}}$$

$$\leq \sum_{n=1}^{\infty} \sum_{ab=n} \frac{|f(a)||g(b)|}{(ab)^{\mathrm{Re}(s)}}$$

$$= \sum_{a,b=1}^{\infty} \frac{|f(a)||g(b)|}{(ab)^{\mathrm{Re}(s)}} = \left(\sum_{a=1}^{\infty} \frac{|f(a)|}{a^{\mathrm{Re}(s)}}\right)\left(\sum_{b=1}^{\infty} \frac{|g(b)|}{b^{\mathrm{Re}(s)}}\right) < \infty.$$

All of the above identities and swaps of order of summation above are justified by the fact that we are summing positive terms. We have thus shown that $\sigma_{f*g} \leq \max(\sigma_f, \sigma_g)$. Moreover, for $\mathrm{Re}(s) > \max(\sigma_f, \sigma_g)$, we have by absolute convergence that we can regroup the terms arbitrarily, and so we have

$$L(s,f)L(s,g) = L(s, f*g).$$

□

## 3.3 Dirichlet series and multiplicative functions

**Definition 3.4.** *Let $f \in \mathcal{A}$ be a multiplicative function of polynomial growth, then for all $\sigma > \sigma_f$ we have*

*For all p prime, the series*

$$L_p(s,f) := \sum_{a \geq 0} \frac{f(p^a)}{p^{as}}$$

*converges absolutely and uniformly in the half plane $\mathrm{Re}(s) \geq \sigma$. We call $L_p(s,f)$ the local factor of $f$ at $p$.*

**Theorem 3.5.**

1. *Moreover, we have*

$$L(s,f) = \prod_p L_p(s,f) = \lim_{P \to \infty} \prod_{p \leq P} L_p(s,f),$$

*and the convergence is uniform in this half-plane.*

2. *More precisely, if we write*

$$L^{>P}(s,f) = \prod_{p > P} L_p(s,f),$$

*then as $P \to \infty$ we have*

$$L^{>P}(s,f) \to 1$$

*uniformly in every half-plane $\mathrm{Re}(s) \geq \sigma$, $\sigma > \sigma_f$.*

3. *Conversely, if $f$ is an arithmetic function such that $\sigma_f < \infty$ and $f(1) = 1$ and if $L(s,f)$ satisfies*

$$L(s,f) = \prod_p L_p(s,f) = \lim_{P \to \infty} \prod_{p \leq P} L_p(s,f)$$

*for $s$ sufficiently large, then $f$ is multiplicative.*

*Proof.*

$$\begin{aligned}
\prod_p L_p(s,f) &= \prod_p \sum_{a\geq 0} \frac{f(p^a)}{(p^a)^s} \\
&= \sum_{a_1\geq 0, a_2\geq 0,\ldots} \frac{f(p_1^{a_1})f(p_2^{a_2})\ldots}{(p_1^{a_1})^s(p_2^{a_2})^s\ldots} \\
&= \sum_{a_1\geq 0, a_2\geq 0,\ldots} \frac{f(p_1^{a_1}p_2^{a_2}\ldots)}{(p_1^{a_1}p_2^{a_2}\ldots)^s} \\
&= \sum_{n\geq 1} \frac{f(n)}{n^s} = L(s,f)
\end{aligned}$$

□

**Corollary 3.6.** *If $f$ is completely multiplicative, then for* $\mathrm{Re}(s) > \sigma_f$ *we have*

$$L(s,f) = \prod_p \left(1 - \frac{f(p)}{p^s}\right)^{-1}.$$

# 4 Primes in Arithmetic Progressions

## 4.1 arithmetic progressions

**Definition 4.1.** *An arithmetic progression is a doubly-infinite* ***subset*** *of* $\mathbb{Z}$ *satisfying the following property: There exists a positive integer* $q > 0$ *such that the distance between two consecutive integers of this subset is always* $q$*. The integer* $q$ *is called the modulus of the arithmetic progression.*

It is easy to see that arithmetic progressions of modulus $q$ are of the form

$$L_{q,a} = a + q\mathbb{Z} \subseteq \mathbb{Z},$$

where $a$ is an integer. We remark that if $a \equiv a' \pmod{q}$, then we have $L_{q,a} = L_{q,a'}$. Thus arithmetic progressions of modulus $q$ are indexed by the congruence classes modulo $q$ (i.e. by the ring $\mathbb{Z}/q\mathbb{Z}$). There are therefore $q$ of them. The integer $a$ is called the class of the arithmetic progression.

**Theorem 4.2** (Dirichlet's theorem on primes in arithmetic progressions)**.** *Let* $a, q > 0$ *be two relatively prime integers. Then, the set*

$$\mathcal{P}_{q,a} = \mathcal{P} \cap L_{q,a}$$

*is infinite. Said differently, there exist infinitely many prime numbers* $p \equiv a \pmod{q}$.

In the vein of the prime number theorem, we can pose more precise questions on the density of the set $\mathscr{P}_{q,a}$. We therefore set

$$\pi(x; q, a) = |\mathscr{P}_{q,a} \cap [1, x]| = |\{p \leq x : p \equiv a \pmod{q}\}| = \sum_{\substack{p \equiv a \pmod{q} \\ p \leq x}} 1$$

the counting function of the primes $p \equiv a \pmod{q}$. At the beginning of the 20th century, Landau showed this generalization of the prime number theorem:

**Theorem 4.3.** *(Landau). Let* $a, q > 0$ *be relatively prime integers. Then*

$$\pi(x; q, a) = \frac{1}{\varphi(q)}\pi(x)(1 + o(1)) = \frac{1}{\varphi(q)}\frac{x}{\log x}(1 + o(1)).$$

*where* $\varphi = \mu * Id$ *is the Euler function,* $\mu = 1^{(-1)}$ *is the Möbius function and* $Id(n) = n$

*is identity function. The Euler function can also be written as the count of numbers between 1 and $n$ that satisfy $(n, q) = 1$.*

Merten's theorem is extended to intersection set of primes and arithmetic progressions.

**Theorem 4.4.** *We have*

$$\sum_{\substack{n \le x \\ n \equiv a \pmod q}} \frac{\Lambda(n)}{n} = \frac{1}{\varphi(q)} \log(x) + O(1) \tag{4.1}$$

$$\sum_{\substack{p \le x \\ p \equiv a \pmod q}} \frac{\log p}{p} = \frac{1}{\varphi(q)} \log(x) + O(1), \tag{4.2}$$

$$\sum_{\substack{p \le x \\ p \equiv a \pmod q}} \frac{1}{p} = \frac{1}{\varphi(q)} \log\log(x) + O(1). \tag{4.3}$$

The crucial point is the group structure of the set $(\mathbb{Z}/q\mathbb{Z})^\times$.

## 4.2 abelian group

The Abelian group is the group in which the group operation is commutative, meaning that the order of operation does not affect the results.

**Definition 4.5.** *Let $G$ be a set of elements and $*$ be an operation. $(G, *)$ is called a* group *if it satisfies the following properties:*

1. ***Closure:*** *For all $a, b \in G$, we have $a * b \in G$.*

2. ***Associativity:*** *For all $a, b, c \in G$,*

$$(a * b) * c = a * (b * c).$$

3. ***Identity element:*** *There exists an element $e \in G$ such that*

$$e * a = a * e = a, \quad \forall a \in G.$$

4. ***Inverse element:*** *For every $a \in G$, there exists an element $a^{-1} \in G$ such that*

$$a * a^{-1} = a^{-1} * a = e.$$

If, in addition, $a * b = b * a$ for all $a, b \in G$, then $G$ is called an *abelian group.*

**Example:**

- $(\mathbb{Z}, +)$ is an abelian group.
- $\{(0,0),(0,1),(1,0),(1,1)\}$ is an abelian group of addition modulo 2.

Let $G$ be a finite abelian group, and $g \in G$ is an element in the group.

**Definition 4.6** (right translation)**.** *The action of* ***right translation*** *is defined as,*

$$T : g \mapsto T_g$$

We can verify that,

$$T_g \circ T_{g'} = T_{gg'} \tag{15}$$

As the action is invertible, the inverse is defined as,

$$(T_g)^{-1} = T_{g^{-1}} \tag{16}$$

As $G$ is abelian group, the action can commute,

$$T_g \circ T_{g'} = T_{g'} \circ T_g \tag{17}$$

The inner product is defined as,

$$\langle f, f' \rangle = \frac{1}{|G|} \sum_{g \in G} f(g)\, \overline{f'(g)}. \tag{18}$$

| **Map Type** | **Domain → Codomain** | **Bijective?** | **Meaning** |
|---|---|---|---|
| Homomorphism | $A \to B$ | Not required | Structure-preserving map |
| Isomorphism | $A \to B$ | Yes | Bijective homomorphism |

**Definition 4.7** (Group homomorphism)**.** *Let* $(G, \circ)$ *and* $(H, *)$ *be groups. A map* $\phi : G \to H$ *is called a* group homomorphism *if*

$$\phi(x \circ y) = \phi(x) * \phi(y) \quad \textit{for all } x, y \in G.$$

**Example:** $\varphi(n) = n \bmod 5$, which $(\mathbb{Z}, +) \mapsto (\mathbb{Z}_5, +)$

**Definition 4.8** (Group character)**.** *Let $G$ be a group. There exists a unique orthonormal basis consisting eigenvectors for all of $T_g$, such that for all $\chi$ we have $\chi(e) = 1$. i.e.*

$$T_g\chi = \chi(g)\chi$$

*A* (group) character *of $G$ is a group homomorphism*

$$\chi : G \longrightarrow \mathbb{C}^\times,$$

*that is, a mapping into the group $\mathbb{C}^\times$ with complex numbers set and the multiplication operation.*

**Example:** Let $G = \mathbb{Z}_3 = \{0, 1, 2\}$ under addition modulo 3. A *character* of $G$ is a homomorphism

$$\chi : (\mathbb{Z}_3, +) \longrightarrow (\mathbb{C}^\times, \cdot)$$

satisfying

$$\chi(a + b) = \chi(a)\chi(b), \quad \forall\, a, b \in \mathbb{Z}_3.$$

The characters of $\mathbb{Z}_3$ are given by

$$\chi_k(a) = e^{2\pi ika/3}, \qquad k = 0, 1, 2.$$

Each $e^{2\pi ika/3}$ lies in $\mathbb{C}^\times$ (since it is nonzero), and

$$\chi_k(a + b) = e^{2\pi ik(a+b)/3} = e^{2\pi ika/3}e^{2\pi ikb/3} = \chi_k(a)\chi_k(b).$$

**Theorem 4.9.** *Let $(V, \langle\,,\,\rangle)$ be a finite-dimensional hermitian vector space, and $\mathcal{T} \subset \mathrm{End}(V)$ a set of pairwise commuting endomorphisms. Then there exists an orthonormal basis of $V$ given by simultaneous eigenvectors of all of the elements of $\mathcal{T}$ if and only if each $T \in \mathcal{T}$ is normal (that is to say, $TT^* = T^*T$).*

In the situation at hand, the spectral theorem implies that $\mathscr{C}(G)$ possesses an orthonormal basis of eigenvectors of all of the $T_g$. In fact, up to permutation, this basis is unique.

**Example:** Fourier Transform defines a unique orthonormal basis $\widehat{G}$. For identity $\delta[n]$, its Fourier transform $\mathcal{F}\{\delta[n]\} = 1$.

**Proposition 4.10** (Orthogonality relations.). *We have*

$$(4.7)\quad \delta_{\chi=\chi'} = \frac{1}{|G|}\sum_{g'\in G}\chi(g')\overline{\chi}(g')$$

*and*

$$(4.8)\quad \delta_{g=g'} = \frac{1}{|G|}\sum_{\chi\in\hat{G}}\overline{\chi}(g)\chi(g').$$

In the case of $\mathbb{Z}/q\mathbb{Z}$, the character is defined as,

$$\psi_n(x) : x \mapsto e^{i2\pi nx/q} \quad x \in \mathbb{Z}/q\mathbb{Z} \tag{19}$$

Note that the character only depends on the class of $n$ modulo $q$.

The trivial character is defined as,

$$\chi_0 : g \mapsto 1 \tag{20}$$

## 4.3 Dirichlet Character

**Definition 4.11** (4.11). *Let $q \geq 1$ be an integer. The characters of the abelian group $(\mathbb{Z}/q\mathbb{Z})^\times$ are called* Dirichlet characters of modulus $q$. *The trivial character (i.e. the constant function equal to* 1*) is denoted $\chi_0$.*

The multiplicative group $(\mathbb{Z}/q\mathbb{Z})^\times$ is defined as the integer between 1 and q that is co-prime with q under multiplication operation. The Dirichlet character of this group is defined as group character *extended* to all integers,

$$\chi(n) = \begin{cases} \chi(n \bmod q), & (n,q) = 1, \\ 0, & (n,q) > 1. \end{cases}$$

which satisfies *completely multiplicative* property (by definition of group homomorphism),

$$\chi(mn) = \chi(m)\chi(n) \text{ for all } m, n \in N_{\geq}$$

We consider Dirichlet series associated with Dirichlet character,

$$L(\chi, s) = \sum_{n\geq 1}\frac{\chi(n)}{n^s} \tag{21}$$

which can be written as,

$$L(\chi, s) = \prod_p (1 - \frac{\chi(p)}{p^s})^{-1} \tag{22}$$

**Proposition 4.12** (4.12)**.** *Let $\chi \pmod q$ be a Dirichlet character.*

- *If $\chi = \chi_0$, we have for* $\mathrm{Re}(s) > 1$

$$L(s, \chi_0) = \prod_{p|q} \left(1 - \frac{1}{p^s}\right) \zeta(s),$$

  *which admits an analytic continuation to the half-plane* $\mathrm{Re}(s) > 0$ *with a simple pole at $s = 1$.*

- *If $\chi \neq \chi_0$, the series $L(s, \chi)$ converges uniformly on compacta in the half-plane* $\mathrm{Re}(s) > 0$ *and thus defines a holomorphic function in this domain. More precisely, we have for* $\mathrm{Re}(s) > 0$

$$\sum_{1 \le n \le X} \frac{\chi(n)}{n^s} = L(s, \chi) + O\left(\frac{q|s|}{\sigma} X^{-\sigma}\right). \tag{4.12}$$

## 4.4 Beginning of the proof of Mertens theorem in arithmetic progressions

To prove Mertens theorem,

$$\sum_{\substack{n \le x \\ n \equiv a \pmod q}} \frac{\Lambda(n)}{n} = \frac{1}{\varphi(q)} \log(x) + O(1) \tag{4.1}$$

Consider the orthogonality, we have,

$$\delta_{\chi = \chi'} = \frac{1}{|G|} \sum_{g \in G} \chi(g) \overline{\chi'(g)} \tag{23}$$

$$\delta_{g = g'} = \frac{1}{\hat{G}} \sum_{\chi \in \hat{G}} \chi(g) \overline{\chi(g')} \tag{24}$$

which are dual relations. For the group $(\mathbb{Z}/q\mathbb{Z})^\times$, we have

$$\delta_{\chi = \chi'} = \frac{1}{\varphi(q)} \sum_{a \bmod q} \chi(a) \overline{\chi'(a)} \tag{25}$$

$$\delta_{a \equiv b \bmod q} = \frac{1}{\varphi(q)} \sum_{\chi \bmod q} \chi(a) \overline{\chi(b)} \tag{26}$$

where $\varphi(q)$ is the Euler function.

we have

$$\sum_{\substack{n\leq x\\ n\equiv a \pmod q}} \frac{\Lambda(n)}{n} = \frac{1}{\varphi(q)} \sum_{\chi \,(\mathrm{mod}\, q)} \overline{\chi}(a) S_\chi(x), \tag{27}$$

where

$$S_\chi(x) = \sum_{n\leq x} \frac{\chi(n)}{n} \Lambda(n).$$

If $\chi = \chi_0$,

$$S_{\chi_0}(x) = \log x + O_q(1)$$

It suffices to prove that the remain terms,

$$S_\chi(x) = O_q(1)$$

The sum of $\chi \neq \chi_0$ can be written as,

$$\sum_{g\in G} \chi(g) = 0$$

as it is periodic,

$$|M_\chi(x)| = |\sum_{n\leq x} \chi(n)| \leq q$$

For the function,

$$T_\chi(x) = \sum_{n\leq x} \frac{\chi(n)}{n} \log(n)$$

by integral of parts

$$T_\chi(x) = \frac{\log(x)}{x} M_\chi(x) + \int_1^x M_\chi(x)(\log n - 1)\frac{1}{n^2} dn$$

is bounded by,

$$T_\chi(x) = O_q(1)$$

On the other hand, $\log = 1 * \Lambda$,

$$\begin{aligned} T_\chi(x) &= \sum_{n\leq x} \frac{\chi(n)}{n} \sum_{ab=n} \Lambda(a) \\ &= \sum_{n\leq x} \frac{\chi(ab)}{ab} \sum_{ab=n} \Lambda(a) \\ &= \sum_{a\leq x} \frac{\chi(a)}{a} \Lambda(a) \sum_{b\leq x/a} \frac{\chi(b)}{b} \\ &= S_\chi(x) L(\chi, 1) \end{aligned}$$

As $L(\chi, 1) \neq 0$, we can prove that,

$$S_\chi(x) = O_q(1)$$

# 5 Riemann's Memoir

More than 150 years ago, in 1859, Riemann published his celebrated memoir *Über die Anzahl der Primzahlen unter einer gegebenen Grösse*. In this memoir, Riemann presented the foundations for the study of the analytic properties of the zeta function

$$\zeta(s) = \sum_{n\geq 1} \frac{1}{n^s} = \prod_p \left(1 - \frac{1}{p^s}\right)^{-1}, \qquad \mathrm{Re}(s) > 1 \tag{28}$$

as a function of a complex variable $s$, and the relation between these and the problem of counting prime numbers.

In his memoir, Riemann mentioned an "explicit" formula, shown later by von Mangoldt, relating the summation function of the function $\Lambda$

$$M_\Lambda(x) = \sum_{n\leq x} \Lambda(n) \tag{29}$$

to a sum over the zeros of $\zeta(s)$. Sufficient information on the location of the zeros of $\zeta(s)$ then implies an asymptotic formula for $M_\Lambda(x)$ as $x \to \infty$.

Riemann showed that there are an infinity of "non-trivial" zeros of $\zeta$, and gave an asymptotic formula for their number. He also formulated in his memoir the celebrated *Riemann hypothesis*, which predicts that all the non-trivial zeros of $\zeta(s)$ are situated on the line

$$\mathrm{Re}(s) = \frac{1}{2}. \tag{30}$$

The proof of this hypothesis is one of the most important problems in mathematics. The Riemann hypothesis is one of the famous Millennium Prize Problems, and the Clay Mathematics Institute has offered a $1,000,000 USD reward for its solution.

# 6 The functional equation

**Definition 6.1.** *Let $f \in \mathcal{S}(\mathbb{R})$. The Fourier transform of $f$ is the function*

$$\hat{f} : y \mapsto \int_{\mathbb{R}} f(x)e(-xy)\, dx. \tag{31}$$

*where* $e(x) = \exp(i2\pi x)$

As $f$ is of rapid decrease, we see that the preceding integral converges absolutely and uniformly on $\mathbb{R}$ and that it defines an infinitely differentiable function with derivatives

$$\hat{f}^{(j)}(y) = \int_{\mathbb{R}} (-2\pi i x)^j f(x)e(-xy)\, dx = \widehat{M^j f}(y), \tag{32}$$

with

$$Mf : x \mapsto (-2\pi i x)f(x). \tag{33}$$

Also note that by integration by parts, we have for all $y \neq 0$ that

$$\hat{f}(y) = \frac{1}{2\pi i y} \int_{\mathbb{R}} f'(x)e(-xy)\, dx = \frac{1}{2\pi i y}\widehat{f'}(y), \tag{34}$$

and iterating this, we obtain for all $j \geq 1$ that

$$\hat{f}(y) = \left(\frac{1}{2\pi i y}\right)^j \widehat{f^{(j)}}(y). \tag{35}$$

## 6.1 Poisson Summation

**Proposition 6.2.** *Let $f \in \mathcal{S}(\mathbb{R})$. For all $u \in \mathbb{R}$ we have*

$$\sum_{n \in \mathbb{Z}} f(n+u) = \sum_{n \in \mathbb{Z}} \hat{f}(n)e(nu). \tag{36}$$

*Proof.* by Fourier series. □

## 6.2 The Mellin transform

**Definition 6.3.** *Let $f \in \mathcal{C}^\infty(\mathbb{R}_{\geq 0})$ be a function that decays rapidly as $x \to \infty$ along with all its derivatives. The Mellin transform of $f$ is defined for* $\mathrm{Re}(s) > 0$ *by*

$$\tilde{f}(s) = \int_{\mathbb{R}_{\geq 0}} f(x) x^s d^\times x, \tag{37}$$

*where we have set*

$$d^\times x = \frac{dx}{x}. \tag{38}$$

It is the measure on $(\mathbb{R}_{>0}, \times)$ which is invariant by translations $x \to yx$.

## 6.3 The Gamma Function

**Definition 6.4.** *The Gamma function* $\Gamma$ *is by definition the Mellin transform of the function* $x \mapsto e^{-x}$*.*

$$\Gamma(s) = \widetilde{e^{-x}}(s) = \int_0^\infty e^{-x} x^s d^\times x.$$

## 6.4 The functional equation of the Riemann zeta function

**Theorem 6.5.** *Let*

$$\xi(s) = \zeta_\infty(s)\zeta(s), \qquad \text{where } \zeta_\infty(s) = \pi^{-s/2}\Gamma(s/2), \qquad \mathrm{Re}(s) > 1. \tag{39}$$

*This function has a meromorphic continuation to* $\mathbb{C}$*, is holomorphic on* $\mathbb{C} \setminus \{0, 1\}$*, and has simple poles at* $s = 0, 1$*. It satisfies the functional equation*

$$\xi(s) = \xi(1 - s). \tag{40}$$

*In particular, the function*

$$\zeta(s) = \sum_{n \geq 1} n^{-s},$$

*defined initially for* $\mathrm{Re}(s) > 1$*, extends to a meromorphic function on* $\mathbb{C}$*, holomorphic in* $\mathbb{C}\setminus\{1\}$*, with a simple pole at* $s = 1$ *with residue* $1$*. It satisfies the functional equation*

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1 - s)\zeta(1 - s). \tag{41}$$

## 6.5 Primitive Characters

Let $\chi$ be a Dirichlet character modulo $q$. There is a natural number associated to $\chi$ called its conductor.

**Definition 6.6** (Conductor)**.** *The minimal $q^\star \mid q$ such that $\chi = \chi_0 \chi^\star$ with $\chi^\star$ a Dirichlet character modulo $q^\star$ and $\chi_0$ the trivial character modulo $q$ is called the conductor of $\chi$ modulo $q$. If $q^\star = q$ then $\chi$ is called primitive.*

Given $\chi$ modulo $q$, the character $\chi^\star$ modulo $q^\star$ in the above factorization is unique, and is called the primitive character inducing $\chi$. Note that we have

$$\chi(a) = \chi^\star(a) \quad \text{if } (a, q) = 1.$$

However, the two characters could take different values if $(a, q^\star) = 1$ but $(a, q) \neq 1$.

# 7 Gauss sums

Let $\chi$ modulo $q$ be a Dirichlet character.

**Definition 7.1** (Gauss sum)**.** *The quantity*

$$\tau(\chi) = \sum_{b \pmod q} \chi(b) e\left(\frac{b}{q}\right)$$

*is called the Gauss sum of $\chi$.*

We have

$$e\left(\frac{a}{q}\right) = \frac{1}{\varphi(q)} \sum_{\chi \pmod q} \overline{\chi}(a) \tau(\chi) \quad \text{if } (a, q) = 1,$$

by the orthogonality relations for Dirichlet characters. The equation above is the Fourier expansion of the additive character as a function on the group $(\mathbb{Z}/q\mathbb{Z})^\times$. Similarly,

$$\chi(a) \tau(\overline{\chi}) = \sum_{b \pmod q} \overline{\chi}(b) e\left(\frac{ab}{q}\right) \quad \text{if } (a, q) = 1.$$